Simpli-Reply

Team: HMJ



Tran Que Anh (Harry) | Nguyen Bui Hoang Tan (Mike) | Nguyen Tuan Anh (Jerry)

RPA x GEN AI HACKATHON 2024

UiPath™ Community

simplifynext

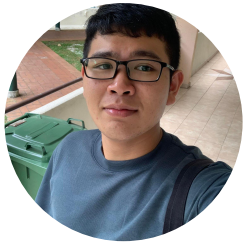
NANYANG TECHNOLOGICAL UNIVERSITY SINGAPORE

Introduction

OUR TEAM



Mike
Electrical and Electronics Engineering/Y3



Harry
Computer Science/Y3



Jerry
Information Engineering and Media/Y3

University Student Inquiry



What are the academic unit requirements to achieve Year 4 status?

- High volume of academic inquiries not addressable via current AI systems.
- Excessive requests burdening university administrative staff.
- Existing AI tools lack capability to resolve specific internal policy queries.

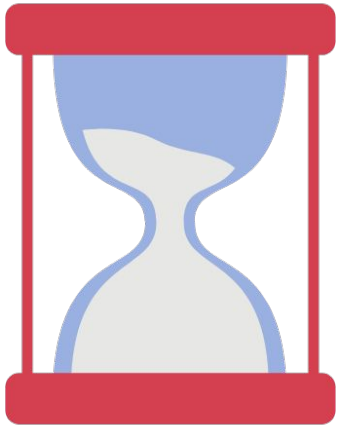
School Officer's Challenges



Typically, it takes 4 to 5 days to respond to student emails.

- Overwhelmed by the multitude of student concerns, leading to email system overcapacity
- Necessity to manage and respond to repetitive query types.
- Automation of the response process would significantly enhance efficiency.

Imperative for Swift and Accurate Customer Service Responses Across Industries



40%

of customers want companies to focus on taking care of their needs quickly

86%

of customers have to contact customer service multiple times for the same reason

Deploying UiPath with embedded Generative AI capabilities can markedly optimize operational efficiency

Source: https://www.netomi.com/customer-service-statistics

Simpli-Reply

Team HMJ

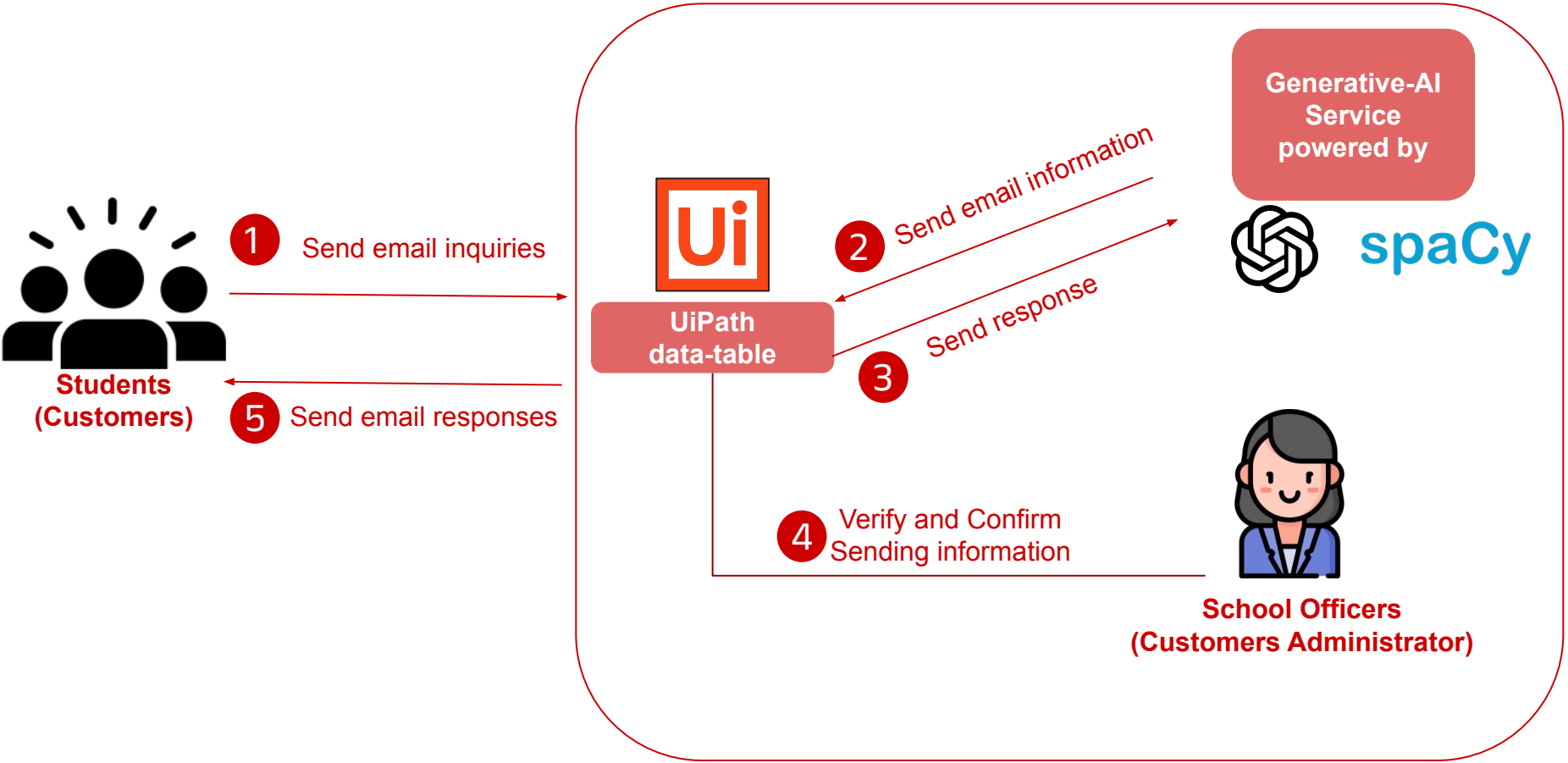
Github Repo

Demo Video

Redefine Your Email Management with AI-Driven RPA Solutions for the Automotive Sector and Beyond

Simpli-Reply utilizes Generative AI to revolutionize response processes, offering educational administrative staff powerful email automation. This technology promises to boost operational efficiency and benefit industries needing superior customer support.

Solution Architecture - How it works?



Unique Selling Point: Retrieval-Augmented Generation Large Language Model (LLM)

Latest GPT models: GPT-4

You
I am Harry, U2120901G. I am studying Computer Science at Nanyang Technological University. How many AUs do I need to complete to be considered a Year-4 student? I could not find this information anywhere.

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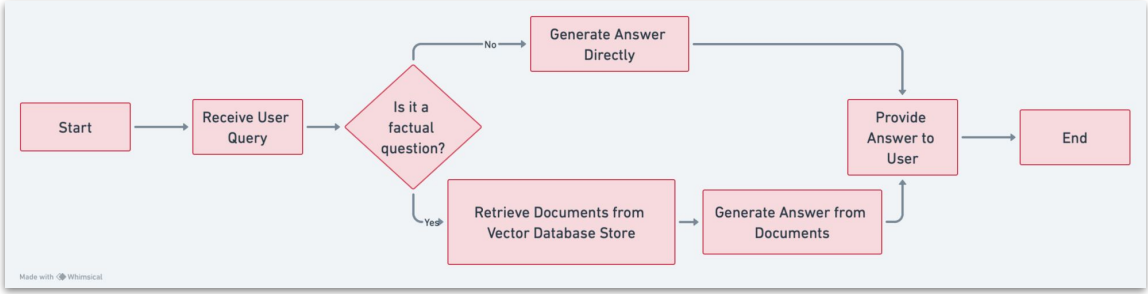
ChatGPT
To find the specific Academic Unit (AU) requirements to be considered a Year-4 student in Computer Science at Nanyang Technological University (NTU), I recommend checking the NTU Academic Structure page directly or the Content of Courses page. Unfortunately, the exact AU requirements were not directly available from the sources accessed. Typically, these requirements vary by program and may be updated annually.

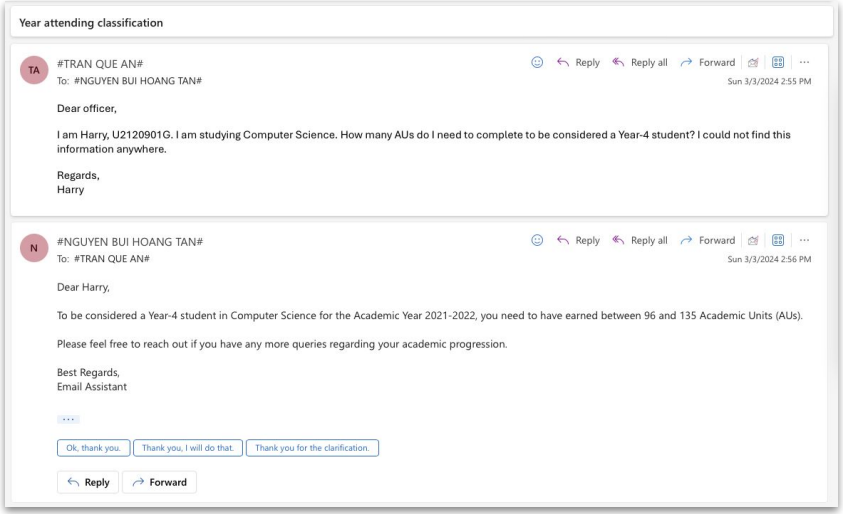
The Academic Structure page on NTU's website and the Content of Courses page are key resources for detailed program requirements, including AUs needed per year, elective options, and core modules. Since the requirements can change, it's best to consult these pages or contact the university directly for the most current information.

For the most accurate and up-to-date information, please refer to NTU's official website or contact the university's academic advising or admissions office directly [1] [2].

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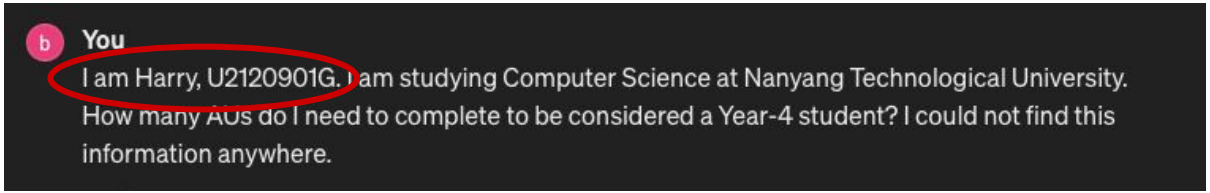
RAG GPT-4 with RAG





Store data in a *vector database* + using *NLP model* to identify Personal Identifiable Information entities and mask them with dummy entities

Which personal information have we exposed to OpenAI?



- **Name**: Harry → **Daniel**
- **Matric number**: U2120901G → **AAOJ9733A**
- **Phone number**: 83017436 → **001-717-778-3084x032**
- **NRIC**: G3701251A → **SLAF1208U**

Technology

End-user	Initially targeting students with a strategy to broaden our customer base
User-department	University administrative personnel
Industries	Starting within the educational sector, with plans to diversify across multiple industries
Uipath product Used	Utilizing UiPath Studio for automation processes
Other technologies	Spacy, GPT-4

Benefits



EFFICIENCY



TIME-SAVING



END-USER EXPERIENCE

Competitor analysis

	Simpli-Reply	Auto-response/ Chatbot service
GenAI + Automation	✓	
Information Privacy	✓	✓
Operational efficiency	✓	
Exact information retrieval	✓	

Future Implementation



Product

- Increase UI and User experience
- Test in different educational administrative services
- Develop a roadmap for long-term feature updates and maintenance



Scalability

- Establish partnerships with academic institutions for research and development support.
- Initiate pilot programs with target customer segments to gather data and refine product features.



Customer Engagement

- Implement feedback loops with users to ensure the product evolves to meet changing needs.
- Launch a beta testing program for active users to contribute to product development.

Thank you!

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